

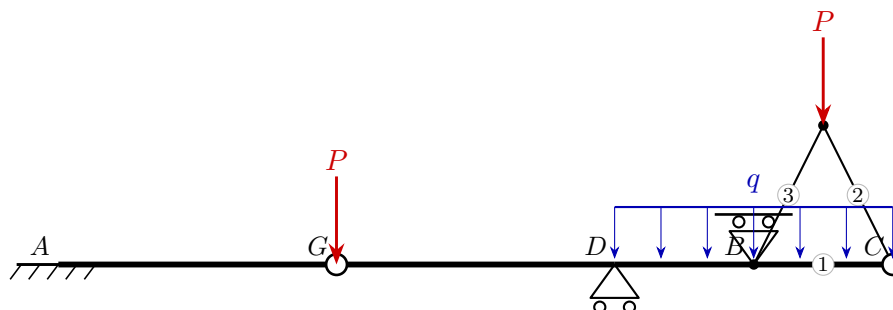
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam Group A

Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 3 - (5 + 4) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

 $A: R_x = 0, R_y = -7.5, M = -22.5 \quad D: R_x = 0, R_y = 37.5 \quad B: R_x = 0, R_y = 7.5$
 $C: H_x = 0, H_y = -7.5 \quad G: H_x = 0, H_y = 15$

3. Truss member forces

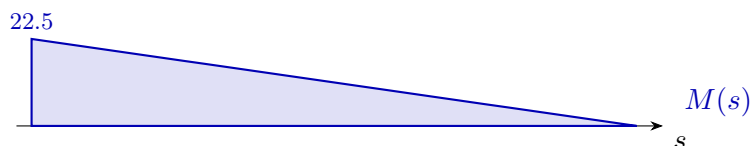
Member	Force S [kN]	Type
1	3.75	tension
2	-8.39	compression
3	-8.39	compression

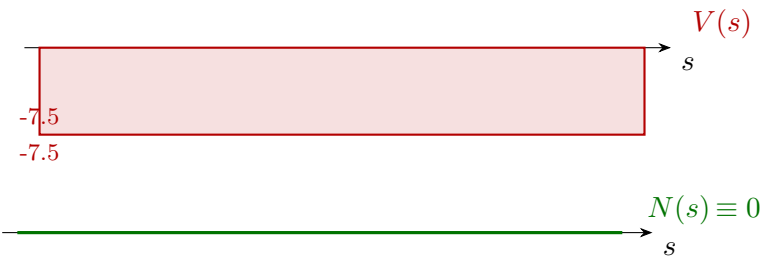
red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams

Beam 1





Beam 2

